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**MINI PROJECT REPORT
ON**

“PORTABLE OSCILLOSCOPE USING ESP32”

Submitted in partial fulfilment of the requirements for the 5th semester of

**BACHELOR OF ENGINEERING
IN
ELECTRONICS AND COMMUNICATION**

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CERTIFICATE

This is to certify that the 5th Semester Mini Project entitled “**PORTABLE OSCILLOSCOPE USING ESP32**” is work carried out by **Suraj Mutagond (2AG23EC041), Rahul Angadi (2AG23EC028), Anmol Ravat (2AG23EC005), Mahammadirfan Yaragatti (2AG23EC016)** in partial fulfillment of the requirements for the award of the degree of **Bachelor of Electronics and communication Engineering under Visvesvaraya Technological University, Belagavi** during the year 2025-2026. It is certified that all the correction/suggestion indicated for internal assessment have been incorporated in the report.

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DECLARATION

We are the students of V semester B.E in Electronics and Communication Engineering, Angadi Institute of Technology and Management, Belagavi, hereby declare that the project work entitled “**PORTABLE OSCILLOSCOPE USING ESP32**” submitted to the Visvesvaraya Technological University during the academic year 2024-25, is a record of an original work done by us under the guidance of **Prof. Pallavi Kulkarni, Assistant Professor Department of Electronics and Communication Engineering**, Angadi Institute of Technology and Management, Belagavi. This project work is submitted in partial fulfilment of the requirements for the award of the degree of Bachelor of Engineering in Electronics and Communication Engineering.

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It is indeed a great moment for us to present our project title “**PORTABLE OSCILLOSCOPE USING ESP32**” However our work never be completed without expressing our gratitude to all the people involved with our project. We take this opportunity to thank them all.

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ABSTRACT

Over the years, oscilloscopes have played a crucial role in the study, analysis, and understanding of electrical and electronic signals. From early analog instruments to today's advanced digital systems, oscilloscopes have continually evolved to meet the needs of engineers, researchers, and students. Signal visualization remains an essential aspect of electronics, enabling the real-time observation of voltage variations, waveform shapes, frequency components, and system behaviour. With the rapid growth of embedded systems and IoT technologies, demand for compact, low-cost, and portable diagnostic tools has increased significantly, especially in educational and field-level applications.

Traditional oscilloscopes, though highly accurate, are often bulky, expensive, and difficult to access for beginners or institutions with limited laboratory resources. These limitations have paved the way for innovative approaches that leverage modern microcontrollers and display technologies to create affordable alternatives. This project presents a comprehensive study and implementation of a **Portable Oscilloscope using ESP32**, aimed at designing a lightweight, efficient, and user-friendly testing instrument capable of measuring and displaying real-time electrical waveforms on an OLED screen.

The system integrates an external high-resolution ADC, an ESP32 microcontroller, and a graphical OLED display to sample, process, and visualize input signals. It offers essential features such as waveform plotting, peak-to-peak voltage measurement, and frequency estimation, making it a valuable tool for students, technicians, and embedded developers. This work also investigates the evolution of portable measurement tools, identifies technological gaps in low-cost oscilloscopes, and highlights the potential for future advancements in the field. The ideas presented in this project contribute to broader efforts in developing accessible electronic diagnostic tools and open new avenues for further research in portable instrumentation and embedded signal processing.

CONTENTS

CHAPTER 1	INTRODUCTION	1
	1.1 Overview of the Proposed System	
	1.2 Motivation	
	1.3 Scope of the Project	
CHAPTER 2	LITERATURE REVIEW	5
	2.1 Review of Existing Work Related to the Project	
CHAPTER 3	PROBLEM DEFINITION AND REQUIREMENT ANALYSIS	6
	3.1 Problem Definition	
	3.2 Objectives	
CHAPTER 4	DESIGN AND IMPLEMENTATION	8
	4.1 System Design	
	4.2 Flow Chart	
	4.3 Methodology	
CHAPTER 5	SOFTWARE AND HARDWARE REQUIREMENTS	11
	5.1 Hardware Requirements	
	5.1.1 ESP32 Microcontroller	
	5.1.2 OLED Display (SSD1306)	
	5.1.3 ADS1115 Analog to Digital Converter (ADC)	
	5.1.4 Jumper Wire	
	5.1.5 Laptop	
	5.2 Software Requirements	
	5.2.1 Arduino IDE	
	5.2.2 ESP32 Code	
CHAPTER 6	TESTING AND RESULTS	23
CHAPTER 7	FUTURE ENHANCEMENTS	26
	7.1 Conclusion	
	7.2 Reference	

LIST OF FIGURES

SL. No	Figure No	Figure Title	Page No
1.	Fig 5.1	ESP32 MICROCONTROLLER	11
2.	Fig 5.2	OLED DISPLAY	12
3.	Fig 5.3	BACK VIEW OF OLED DISPLAY MODULE	13
4.	Fig 5.4	PIN DESCRIPTION OF OLED DISPLAY MODULE	13
5.	Fig 5.5	ADS1115 ADC	15
6.	Fig 5.6	JUMPER WIRES	16
7.	Fig 5.7	LAPTOP	17
8.	Fig 6.1	REAL-TIME WAVEFORM DISPLAYED ON OLED	24
9.	Fig 6.2	HARDWARE SETUP OF THE PORTABLE OSCILLOSCOPE	24
10.	Fig 6.3	GROUP PHOTO WITH GUIDE	25

CHAPTER - 1

INTRODUCTION

Oscilloscopes are fundamental tools in the field of electronics, used to visualize, measure, and analyze electrical signals. They enable engineers, researchers, and students to observe waveform behavior, detect faults, verify circuit performance, and study system responses in real time. Over the years, oscilloscopes have evolved from bulky analog instruments to advanced digital systems capable of high-speed sampling and complex waveform processing. However, despite these developments, traditional oscilloscopes remain costly, large, and often inaccessible to beginners, small laboratories, and field technicians.

With the rapid growth of embedded systems and low-power microcontrollers, it has become possible to develop compact and affordable alternatives that can perform essential signal-monitoring tasks. The **Portable Oscilloscope using ESP32** is one such solution that aims to provide basic waveform visualization, frequency analysis, and voltage measurement in a compact, low-cost, and user-friendly form. This project leverages the processing capability of the ESP32 microcontroller, the precision of an external ADC (ADS1115), and the graphical output of an OLED display to create a simple yet effective instrument suitable for educational and field applications.

1.1 OVERVIEW OF THE PROPOSED SYSTEM :

The proposed system is a handheld, microcontroller-based oscilloscope capable of capturing and displaying electrical waveforms in real time. It consists of four major components: a signal input stage, an external high-resolution ADC (ADS1115), the ESP32 microcontroller, and a compact OLED display. The input analog signal is first conditioned and fed into the ADC, where it is sampled and converted into digital values. These values are then processed by the ESP32, which reconstructs the waveform, calculates important signal parameters such as frequency and peak-to-peak voltage (V_{pp}), and sends the processed data to the OLED screen for visualization. The system

operates at moderate sampling speeds, making it suitable for low- to mid-frequency signals typically required in hobby, academic, and introductory electronic applications.

This design provides a simplified oscilloscope interface where the user can observe waveform shape, amplitude variation, and other characteristics without the need for a full laboratory oscilloscope. The compact size and low cost make it ideal for portable use, debugging small circuits, and learning signal analysis fundamentals.

1.2 MOTIVATION :

The motivation for developing a portable oscilloscope arises from the increasing need for accessible, real-time, and cost-effective tools for analyzing electrical signals in both academic and practical environments. Oscilloscopes play a fundamental role in understanding circuit behavior, debugging electronic systems, validating sensor outputs, and studying waveform characteristics. However, traditional benchtop oscilloscopes are often expensive, bulky, and limited to laboratory settings, making them difficult to access for students, hobbyists, and field technicians. This creates a significant barrier for individuals who require continuous exposure to signal-analysis tools to enhance their learning and troubleshooting capabilities.

In recent years, advancements in embedded systems and microcontroller technology have transformed the landscape of electronic instrumentation. Devices such as ESP32, high-resolution ADCs, and compact OLED displays now make it possible to design lightweight and affordable oscilloscopes without compromising essential functionalities. These innovations enable users to visualize waveforms, measure signal parameters, and perform basic diagnostic tasks outside of traditional laboratory environments. The availability of such portable tools empowers learners and engineers to carry out experimentation and circuit testing anytime and anywhere, thereby strengthening their practical understanding of electronics.

Furthermore, the increasing adoption of IoT devices, embedded systems, and digital electronics in modern engineering education has amplified the need for portable, easy-to-use diagnostic instruments. Wireless communication, low-power operation, and compact hardware platforms allow for the creation of handheld oscilloscopes that are not only economical but also highly convenient for on-site testing, prototyping, and

classroom demonstrations. These tools help bridge the gap between theoretical knowledge and real-world application, encouraging hands-on learning and innovation.

The motivation behind this project is therefore rooted in the desire to create an efficient, user-friendly, and affordable signal-monitoring system that expands access to essential electronic measurement tools. By leveraging modern microcontrollers and display technologies, this project aims to support students, educators, and electronics practitioners in developing a deeper understanding of signal behavior while reducing dependency on costly laboratory equipment.

1.3 SCOPE OF THE PROJECT :

The scope of this project focuses on the development of a compact, low-cost, and portable oscilloscope designed to support basic electrical signal analysis in both academic and field environments. This portable oscilloscope leverages the ESP32 microcontroller, a high-resolution external ADC, and an OLED display to capture, process, and visualize real-time waveforms. The system is intended to serve as an accessible diagnostic tool for students, hobbyists, technicians, and educators who require essential signal-monitoring capabilities without relying on bulky or expensive laboratory-grade oscilloscopes.

The project includes hardware design, ADC integration, ESP32 firmware development, sampling and signal-processing techniques, and the graphical representation of waveforms. It supports the measurement of low-to-moderate frequency signals, along with the calculation of key parameters such as peak-to-peak voltage, amplitude variation, and frequency estimation. The device is developed with emphasis on portability, ease of use, and affordability, ensuring that users can conveniently analyze basic circuit behavior and troubleshoot electronic components in real time.

The project scope also extends to exploring the limitations of microcontroller-based oscilloscopes and identifying opportunities for future enhancements. Although the device does not aim to replace full-fledged laboratory oscilloscopes, it lays the foundation for further improvements such as multi-channel input, waveform storage, advanced filtering techniques, higher sampling rates, and wireless data transmission to external devices.

By defining clear functional boundaries and providing a practical implementation, this project contributes to the growing field of portable electronic instrumentation and supports hands-on learning in embedded systems and signal analysis.

CHAPTER - 2

LITERATURE REVIEW

2.1 REVIEW OF EXISTING WORK RELATED TO THE PROJECT :

Research in portable measurement instruments has evolved significantly with the advancement of embedded hardware. Several authors have explored the development of low-cost oscilloscopes, microcontroller-based waveform analyzers, and portable diagnostic systems. These studies highlight the increasing need for compact signal monitoring tools suitable for academic and field applications. A summarized review of relevant existing work is given in Table 5.1.

Author	Name of Journal	Year	Proposed Method	Performance Parameter	Advantages	Limitations
A. Vora, R. Patel	Low-Cost Digital Oscilloscope Using Arduino	2019	Designed a basic oscilloscope using Arduino UNO and TFT display	Sampling rate & display clarity	Low-cost, suitable for beginners	Very low sampling rate; cannot analyze higher-frequency signals
K. Sharma, P. Bhatia	Portable Embedded Oscilloscope System	2020	Used ARM Cortex-M microcontroller with graphical LCD to plot signals	Bandwidth, Vpp accuracy	Portable, battery-operated	Expensive MCU, higher complexity
S. Anand, N. Kumar	ESP32-Based Signal Analyzer	2021	Implemented waveform sampling using ESP32 with inbuilt ADC	ADC resolution & refresh rate	Wireless connectivity, IoT-ready	ESP32 ADC not accurate for analog precision
M. Deshmukh	ADS1115 for High-Precision Measurements	2022	Proposed external ADC integration for improved signal accuracy	Bit-resolution, noise reduction	Highly accurate readings	Limited maximum sampling speed
R. Mehta	OLED-Based Microcontroller Oscilloscope	2023	Displayed waveforms on SSD1306 OLED using 8-bit MCU	Display resolution & signal readability	Very compact display system	Screen size restricts visibility

CHAPTER - 3

PROBLEM DEFINITION AND REQUIREMENT ANALYSIS

3.1 PROBLEM DEFINITION :

Oscilloscopes are essential tools for observing and analysing the behaviour of electrical signals in electronic circuits. They allow users to monitor waveform shapes, amplitude variations, frequency changes, and transient responses. However, traditional laboratory oscilloscopes are often large, expensive, and not easily accessible to students, hobbyists, or technicians working in the field. Their high cost and limited portability create a significant barrier for those who require frequent exposure to signal-monitoring tools for learning, experimentation, and troubleshooting.

Although several low-cost solutions exist, many suffer from critical limitations such as low sampling rates, poor ADC resolution, noisy measurements, or lack of real time graphical interfaces. Microcontrollers with built-in ADCs often fail to provide the accuracy needed for reliable waveform reconstruction. Additionally, the absence of compact display units makes most DIY solutions difficult to interpret, reducing their practical value.

Therefore, there is a clear need for a **compact, affordable, and user-friendly portable oscilloscope** that supports real-time waveform visualization and basic signal analysis. Such a system must incorporate an external high-resolution ADC to ensure accuracy, a capable microcontroller to process sampled data, and a compact graphical interface to display waveforms effectively. The solution must be lightweight, easy to use, and suitable for basic signal testing, educational demonstrations, and small-scale circuit diagnostics.

3.2 OBJECTIVES :**1. Develop a Portable Oscilloscope System:**

To design and implement a compact oscilloscope using the ESP32 microcontroller, ADS1115 ADC, and OLED display for real-time measurement and visualization of electrical waveforms.

2. Achieve Accurate Signal Sampling:

To integrate a high-resolution external ADC to enhance sampling accuracy, reduce noise, and improve waveform reconstruction for low-to-moderate frequency signals.

3. Enable Real-Time Waveform Display:

To process sampled data on the ESP32 and plot the waveform on an OLED screen, allowing users to observe voltage variations and waveform shapes instantly.

4. Measure Essential Signal Parameters:

To compute and display key characteristics such as peak-to-peak voltage (V_{pp}), amplitude, and frequency, providing users with meaningful diagnostic information.

5. Ensure Portability and Ease of Use:

To develop a low-cost, lightweight, and battery-operated solution that can serve students, technicians, and hobbyists in both classroom and field environments.

6. Improve Accessibility to Test Equipment:

To deliver an affordable alternative to traditional oscilloscopes, making basic signal analysis tools more accessible for educational institutions and individual learners.

CHAPTER - 4

DESIGN AND IMPEMETATION

4.1 SYSTEM DESIGN :

The system design of the Portable Oscilloscope using ESP32 centers around accurately capturing analog electrical signals, converting them into digital samples, processing them in real time, and visually displaying the resulting waveform. The architecture is composed of four major functional blocks: the signal input stage, analog-to-digital conversion, digital processing, and graphical display.

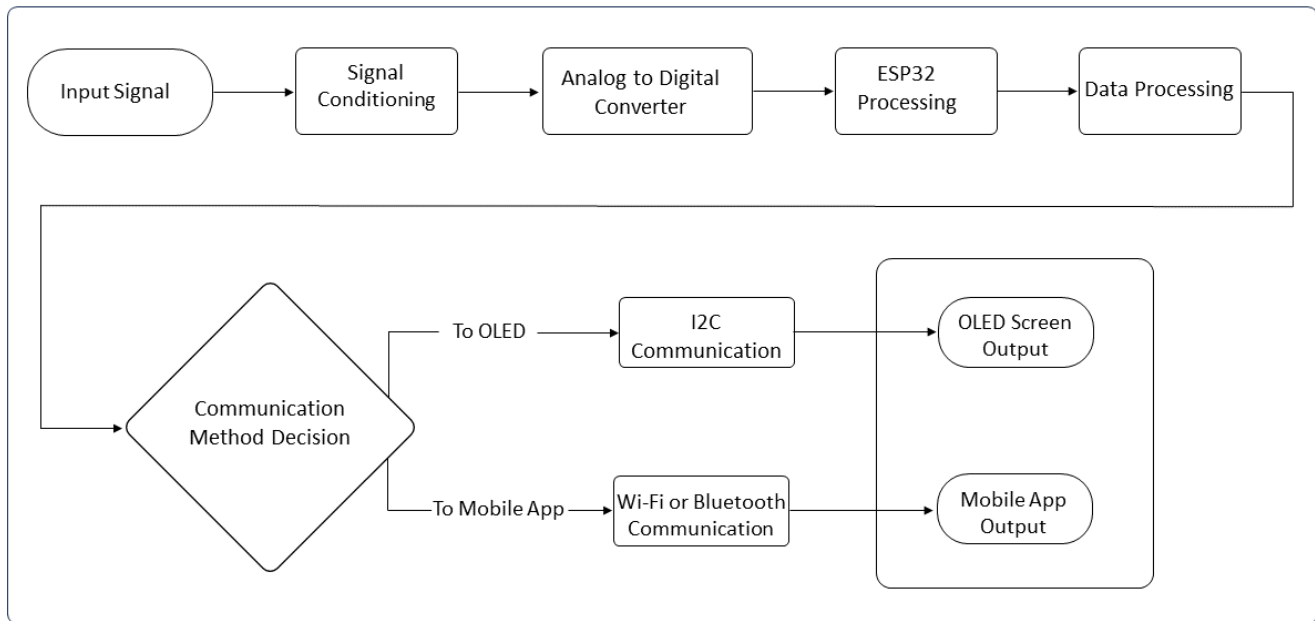
The input electrical signal is first fed into the **ADS1115 external ADC**, which provides high-resolution 16-bit sampling. This is necessary because the built-in ADC of the ESP32 often suffers from noise and non-linearity, making it unsuitable for precision waveform analysis. The ADC samples the input signal at regular intervals and generates digital values representing instantaneous voltages.

These digital samples are transmitted to the **ESP32 Microcontroller**, which serves as the core processing unit. The ESP32 executes several operations, including buffering the incoming data, scaling the sampled values, calculating signal parameters such as peak-to-peak voltage (V_{pp}) and frequency, and reconstructing the waveform for display. Its high processing speed and dual-core architecture allow smooth real-time graph updates.

The processed data is then rendered on a **128×64 OLED display**, where the waveform is plotted graphically across the screen. The display updates continuously, allowing the user to observe changes in the signal in real time. In addition, textual information such as frequency, V_{pp} , amplitude, and sampling rate can be shown on the screen.

The entire system is designed to be compact, power-efficient, and suitable for use in academic laboratories, prototyping environments, and field-level diagnostics. The modular design also allows future expansion, such as adding wireless communication, multiple channels, or data storage.

4.2 FLOW CHART :



4.3 METHODOLOGY :

Step 1: Hardware Setup:

- The input signal is connected to the ADS1115 external ADC.
- The ESP32 communicates with the ADC through the I2C protocol.
- The OLED display is also interfaced using I2C, ensuring synchronized data flow.
- A regulated 3.3V supply powers the microcontroller and display modules.

Step 2: ADC Sampling:

- The ADS1115 samples the analog input with high precision (16-bit).
- Sampling intervals are set to maintain stable waveform capture for low-to-mid frequencies.
- The raw digital output is transferred to the ESP32.

Step 3: Signal Processing:

- The ESP32 converts raw ADC values into measurable voltage units.
- A buffer array stores sequential samples to reconstruct the waveform.
- Algorithm's compute:
 - Peak-to-peak voltage

- Frequency through zero-crossing or time-period detection
- Amplitude and signal shape

Step 4: Display Rendering:

- The OLED display updates at a defined refresh rate.
- The waveform is drawn using pixel mapping, ensuring smooth transitions.
- Numerical values like frequency and V_{pp} are shown alongside the waveform.

Step 5: Iterative Looping:

- The system continuously repeats sampling and processing to maintain real-time visualization.
- Any change in input signal (frequency, amplitude, noise) is instantly reflected on the OLED.

Step 6: Testing & Calibration:

- The oscilloscope is tested using known waveforms from a signal generator.
- Calibration ensures accurate amplitude scaling and frequency detection.

5.1 HARDWARE REQUIREMENT :

5.1.1 ESP32 MICROCONTROLLER :

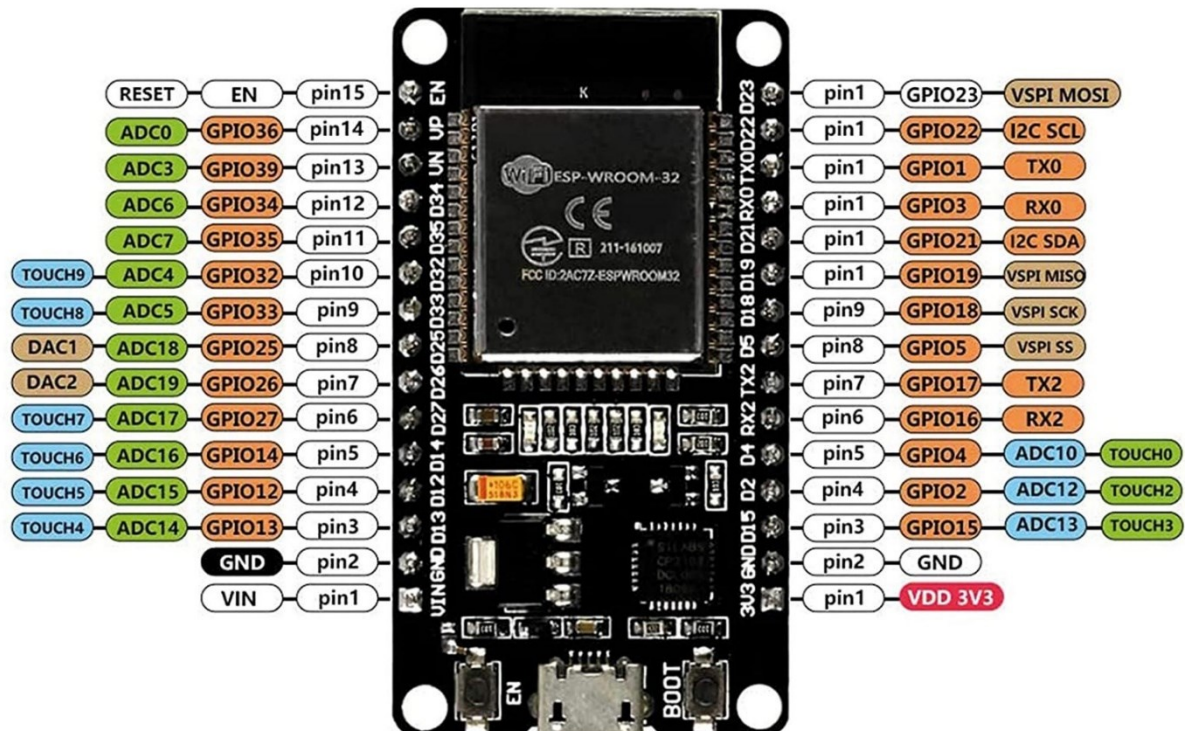


Fig 5.1:ESP32 MICROCONTROLLER

The ESP32 is a powerful dual-core microcontroller widely used in embedded system applications due to its high processing speed, built-in Wi-Fi/Bluetooth capability, and extensive peripheral support. Unlike traditional microcontrollers that offer limited computational performance, the ESP32 integrates a 32-bit Tensilica LX6 processor capable of operating at up to 240 MHz, making it suitable for real-time signal processing required in oscilloscope applications.

The board features multiple GPIO pins, I2C, SPI, UART communication interfaces, and supports both analog and digital peripherals. Although the ESP32 includes internal ADC channels, they suffer from noise and non-linearity, making them unsuitable for high-precision measurements. Therefore, this project uses the

ESP32 primarily for waveform calculation, display handling, digital communication, and overall system control.

The ESP32 can be powered either through a USB connection or via an external 5V source, providing flexibility during prototyping and portable use. It includes voltage regulators, an antenna, and flash memory, enabling it to function as a standalone module without requiring additional support circuitry. Due to its low-power features and deep-sleep modes, the ESP32 is also well suited for portable oscilloscope designs, where efficient energy usage is essential.

5.1.2 OLED DISPLAY (SSD1306) :

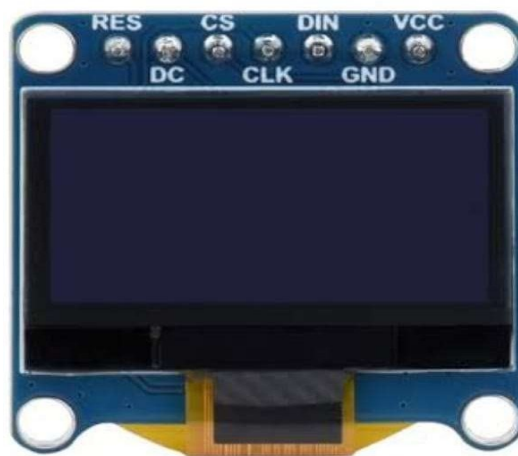
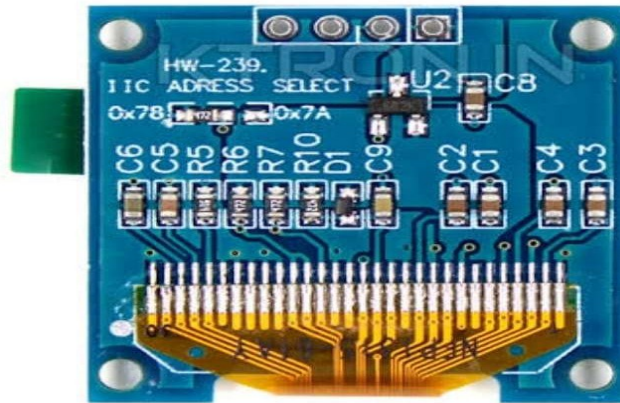
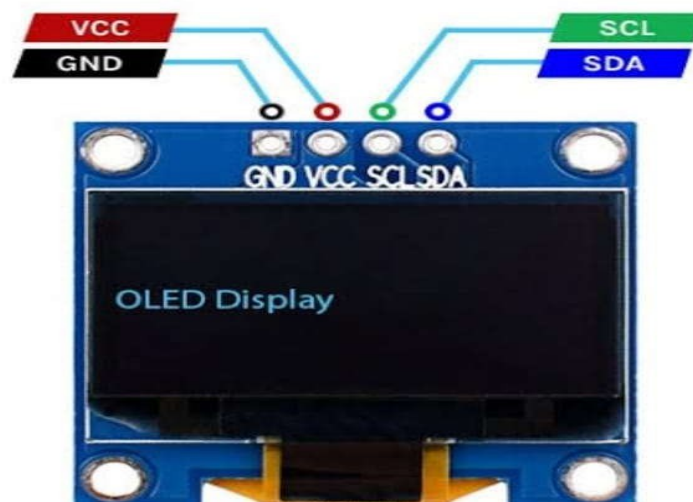


Fig 5.2: OLED DISPLAY

In Fig.3, An OLED (Organic Light Emitting Diode) display is a type of flat-panel display technology that uses organic compounds to emit light when an electric current passes through them. Unlike traditional LCD screens, OLED displays do not require a backlight, as each pixel generates its own light. This allows for better contrast ratios, deeper blacks, and more vibrant colors. OLED displays are also thinner, more flexible, and energy-efficient, making them ideal for portable devices like smartphones, wearables, and TVs. Additionally, OLED screens offer faster response times and wide viewing angles, providing superior visual quality. In addition to their thin form factor and energy efficiency, OLED displays also provide faster response times, which results in smoother motion when viewing videos or playing games. The technology also offers wide viewing angles, maintaining consistent image quality regardless of the viewer's position relative to the screen. OLED displays are also used in larger formats, such as televisions, where their superior contrast ratios and vibrant.

OLED OPERATION :*Fig 5.3: BACK VIEW OF OLED DISPLAY MODULE*

OLED (Organic Light Emitting Diode) operates by using organic compounds that emit light when an electric current passes through them. The OLED structure consists of several layers: a substrate, an anode, organic layers (conductive and emissive), and a cathode. When voltage is applied, the anode extracts positive charge carriers (holes), while the cathode injects electrons into the emissive layer. When these electrons and holes combine, they form excitons, which release energy in the form of light. The color of the emitted light depends on the materials used in the emissive layer.

PIN DESCRIPTION :*Fig 5.4: PIN DESCRIPTION OF OLED DISPLAY MODULE*

OLED pin description typically refers to the pins and their functions in a display module or OLED panel used in electronic circuits. The exact pin configuration may vary depending on the specific OLED display model, but common OLED displays (such as those based on the SSD1306 controller) generally have the following pins and functions:

1. VCC (Power Supply Pin)

Function: Provides power to the OLED display.

Voltage: Typically, 3.3V or 5V, depending on the display.

2. GND (Ground Pin)

Function: Ground pin for the display, completing the electrical circuit.

3. SCL (Serial Clock Line)

Function: Used for serial communication (I2C or SPI). It carries the clock signal for timing data transmission. I2C Communication: SCL is the clock signal used to synchronize data transfer.

4. SDA (Serial Data Line)

Function: Used for serial data transfer. It carries the data to be sent to the OLED display. I2C Communication: SDA is the data line used to send information to the display.

5. RES (Reset Pin)

Function: Resets the OLED display to its default state. This pin is typically connected to a microcontroller to trigger a reset sequence.

6. DC (Data/Command Pin)

Function: Determines whether the data being sent is command data or display data. LOW: Command mode (used to send commands to the display).

HIGH: Data mode (used to send pixel data to the display).

7. CS (Chip Select Pin)

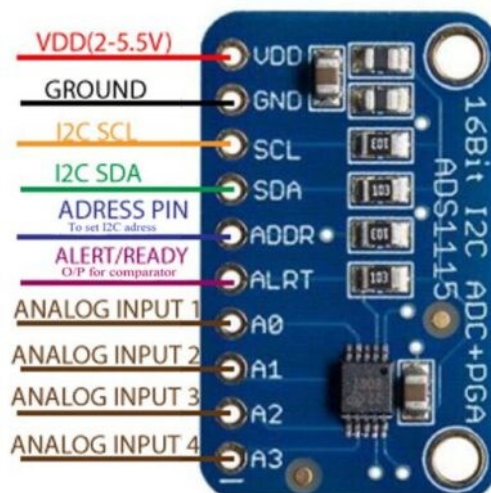
Function: Used in SPI communication to select the OLED module. It must be set low to activate the OLED and allow communication.

8. VCC (Vbat) (Optional for some displays)

Function: A power supply pin that may be used for certain OLEDs with additional functionality, like integrated battery-backed memory.

PIN DESCRIPTION FOR OLED :

Item	Signal	Symbol	Min.	Typ.	Max.	Unit	Remark
Address setup time	RS	t_{AS68}	20	-	-	ns	
Address hold time		t_{AH68}	0	-	-	ns	
System cycle time		t_{CY68}	500	-	-	ns	
Pulse width (E)	E_RDB	$t_{PW68}(W)$	250	-	-	ns	
Pulse width (E)	E_RDB	$t_{PW68}(R)$	250	-	-	ns	
Data setup time	DB7 to DB0	t_{DS68}	40	-	-	ns	
Data hold time		t_{DH68}	20	-	-	ns	
Read access time	DB0	t_{ACC68}	-	-	180	ns	$C_L = 100pF$
Output disable time		t_{OD68}	10	-	-	ns	

5.1.3 ADS1115 ANALOG – TO – DIGITAL CONVERTER (ADC) :*Fig 5.5: ADS1115 ADC*

The ADS1115 is a high-resolution, 16-bit analog-to-digital converter used to capture electrical waveforms with greater precision than the internal ADC of the ESP32. It supports programmable gain amplification, which allows the device to measure small signals accurately by adjusting input voltage ranges. The module communicates with the ESP32 via the I2C protocol, ensuring fast and reliable data transfer.

The ADC samples input signals at selectable sampling rates, making it suitable for low- to mid-frequency waveform acquisition required in a portable oscilloscope. Its high precision ensures minimal distortion and noise, allowing clear visualization on the OLED display. The ADS1115 contains four analog channels, enabling multi-input expansion in future upgrades.

This module plays a crucial role in improving the oscilloscope's measurement accuracy, providing stable digital output for further processing by the ESP32.

5.1.4 JUMPER WIRES :



Fig 5.6: Jumper wires

The Fig.4, jumper wires are essential for establishing electrical connections between the various components, such as the Arduino Uno, sensors LCD display, and power supply. They provide a flexible and quick way to connect pins on the Arduino to external components without soldering, making it easier to build, test, and modify the circuit during prototyping. Jumper wires come in different types (male-to-male, male-to-female, and female-to-female). They are used according to the needs of the circuit, ensuring reliable signal transmission and power distribution.

5.1.5 LAPTOP

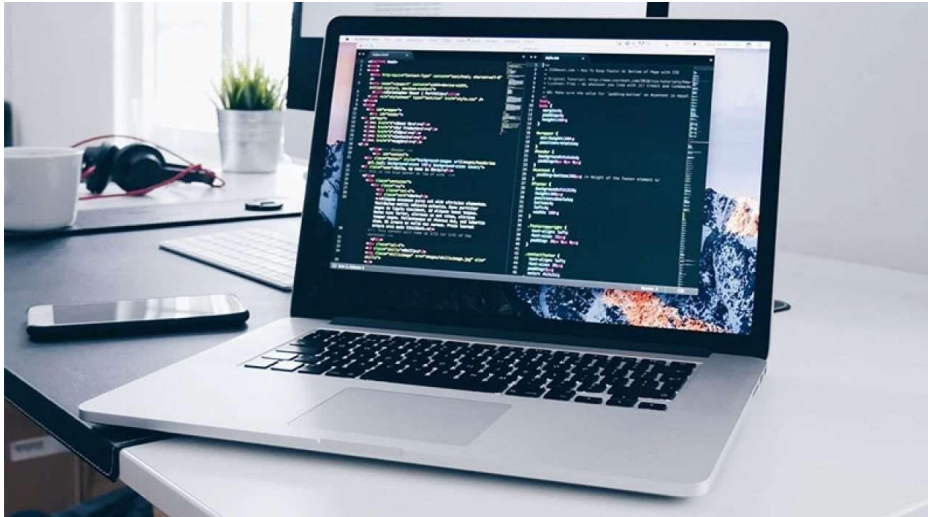


Fig 5.7: Laptop

The laptop plays a crucial role primarily during the development and testing phases. It is used to program the Arduino Uno via the Arduino Integrated Development Environment (IDE). Using a USB cable, the laptop allows the user to write, compile, and upload the control code. Additionally, it serves as a tool for real-time debugging and monitoring through the serial monitor feature of the Arduino IDE, where sensor data can be observed and analyzed. The laptop also provides power to the Arduino during testing. It can be used to make adjustments to the code efficiently, enabling iterative improvements and troubleshooting throughout the development of the adaptive matrix headlight system.

5.2 SOFTWARE REQUIREMENT :

Software plays a crucial role in ensuring that the portable oscilloscope operates efficiently, accurately, and in real time. The development environment, programming tools, and supporting libraries collectively enable the ESP32 to interface with the ADS1115 ADC, process sampled data, compute waveform parameters, and visualize the results on the OLED display. This section details the software platforms and tools required for the successful implementation of the system.

5.2.1 Arduino IDE :

The Arduino Integrated Development Environment (IDE) is the primary software tool used for writing, compiling, and uploading the program to the ESP32 microcontroller. It provides a user-friendly interface, making it suitable for both beginners and experienced developers. The IDE supports multiple microcontroller platforms through board packages, including ESP32, enabling seamless integration during code development.

The Arduino IDE includes an inbuilt text editor for writing code, a compiler for converting high-level instructions into machine-readable format, and an upload module for transferring the compiled program to the ESP32 via USB. The Serial Monitor and Serial Plotter tools within the IDE are valuable during debugging and testing. They allow real-time observation of ADC samples, signal values, and communication responses, helping developers verify waveform accuracy and timing behavior.

To support the portable oscilloscope project, additional libraries are installed through the Library Manager in Arduino IDE, such as the **Adafruit GFX**, **Adafruit SSD1306**, and **Adafruit ADS1X15** libraries. These libraries simplify tasks like drawing waveforms, interfacing with the OLED display, and reading precise voltage samples from the ADS1115 ADC.

The IDE also manages dependencies, ensures compatibility with the selected ESP32 board version, and provides error messages and warnings to assist the developer in refining the code. Due to its simplicity, cross-platform support (Windows, macOS, Linux), and strong community backing, the Arduino IDE is an ideal environment for developing embedded applications like the portable oscilloscope.

5.2.3 ESP32 CODE :

```

#include <Wire.h>
#include <Adafruit_ADS1X15.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>
#include <math.h>

#define SCREEN_WIDTH 128
#define SCREEN_HEIGHT 64
#define OLED_RESET -1
#define OLED_ADDR 0x3C
#define BTN_PIN 32
#define SDA_PIN 21
#define SCL_PIN 22

Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire,
OLED_RESET);
Adafruit_ADS1115 ads;

#define NUM_SAMPLES 128
#define VREF 4.096
#define LSB (VREF / 32767.0)

int16_t samples[NUM_SAMPLES];
unsigned long timestamps[NUM_SAMPLES];
int bufIndex = 0;
bool paused = false;
bool lastButtonState = HIGH;
unsigned long lastDisplayMs = 0;
const unsigned long DISPLAY_INTERVAL = 40;

void setup() {
  Serial.begin(115200);
  Wire.begin(SDA_PIN, SCL_PIN);
  pinMode(BTN_PIN, INPUT_PULLUP);
  if (!display.begin(SSD1306_SWITCHCAPVCC, OLED_ADDR)) while (1);
  if (!ads.begin(0x48)) while (1);
  ads.setGain(GAIN_ONE);
  ads.setDataRate(RATE_ADS1115_860SPS);
  display.clearDisplay();
  display.setTextColor(SSD1306_WHITE);
  display.setTextSize(1);
  display.setCursor(10, 25);
  display.println("Portable Oscilloscope");
  display.display();
  delay(1000);
  for (int i = 0; i < NUM_SAMPLES; ++i) {
    samples[i] = 0;
  }
}

```

```

    timestamps[i] = micros();
  }
}

void loop() {
  handleButton();
  if (!paused) takeSample();
  if (millis() - lastDisplayMs > DISPLAY_INTERVAL) {
    drawScope();
    lastDisplayMs = millis();
  }
}

void handleButton() {
  bool state = digitalRead(BTN_PIN);
  if (!state && lastButtonState) {
    paused = !paused;
    delay(180);
  }
  lastButtonState = state;
}

void takeSample() {
  static unsigned long lastMicros = 0;
  unsigned long now = micros();
  if (now - lastMicros < 1200) return;
  lastMicros = now;
  long sum = 0;
  for (int i = 0; i < 4; i++) sum += ads.readADC_SingleEnded(0);
  samples[bufIndex] = (int16_t)(sum / 4);
  timestamps[bufIndex] = now;
  bufIndex = (bufIndex + 1) % NUM_SAMPLES;
}

float computeSampleRate() {
  unsigned long newest = timestamps[(bufIndex + NUM_SAMPLES - 1) %
NUM_SAMPLES];
  int oldestIndex = bufIndex;
  unsigned long oldest = timestamps[oldestIndex];
  unsigned long dt = (newest >= oldest) ? (newest - oldest) : (ULONG_MAX - oldest +
newest);
  if (dt == 0) return 0;
  return (float)(NUM_SAMPLES - 1) * 1e6f / (float)dt;
}

float estimateFrequencyFromZeroCrossings(int16_t minVal, int16_t maxVal) {
  float mid = (minVal + maxVal) * 0.5f;
  const int maxCrossings = NUM_SAMPLES / 2;
  float crossingTimes[maxCrossings]; // in seconds
  int crossCount = 0;

```

```

int start = bufIndex; // newest index is bufIndex - 1, oldest is bufIndex
int prevIdx = (start + NUM_SAMPLES - 1) % NUM_SAMPLES;
int curIdx;

for (int i = 0; i < NUM_SAMPLES; ++i) {
    curIdx = (start + i) % NUM_SAMPLES;
    int16_t v1 = samples[prevIdx];
    int16_t v2 = samples[curIdx];
    unsigned long t1 = timestamps[prevIdx];
    unsigned long t2 = timestamps[curIdx];

    if ((v1 <= mid) && (v2 > mid)) {
        if (v2 != v1) {
            float frac = (mid - v1) / (float)(v2 - v1);
            float t_cross_s = (t1 + frac * (float)(t2 - t1)) * 1e-6f;
            if (crossCount < maxCrossings) crossingTimes[crossCount++] = t_cross_s;
        }
    }
    prevIdx = curIdx;
}

if (crossCount < 2) return 0.0f;

double sumPeriods = 0.0;
int periodsFound = 0;
for (int k = 1; k < crossCount; ++k) {
    double period = crossingTimes[k] - crossingTimes[k - 1];
    if (period > 0.0005 && period < 1.0) { // ignore absurd values (keep 0.5ms..1s)
        sumPeriods += period;
        periodsFound++;
    }
}
if (periodsFound == 0) return 0.0f;
double avgPeriod = sumPeriods / periodsFound;
if (avgPeriod <= 0.0) return 0.0f;
return (float)(1.0 / avgPeriod);
}

void drawScope() {
    int16_t minVal = samples[0], maxVal = samples[0];
    for (int i = 1; i < NUM_SAMPLES; i++) {
        if (samples[i] < minVal) minVal = samples[i];
        if (samples[i] > maxVal) maxVal = samples[i];
    }

    float mid = (minVal + maxVal) * 0.5f;
    float vpp = (maxVal - minVal) * LSB;
    double sumSq = 0.0;
    for (int i = 0; i < NUM_SAMPLES; i++) {

```

```

    float v = (samples[i] - mid) * LSB;
    sumSq += v * v;
}
float vrms = sqrt(sumSq / NUM_SAMPLES);
float dc = mid * LSB;

float sampleRate = computeSampleRate();
float freq = estimateFrequencyFromZeroCrossings(minVal, maxVal);

display.clearDisplay();
display.setTextSize(1);
display.setCursor(0, 0);
display.printf("Vpp: %.2fV  Vrms: %.2fV", vpp, vrms);

int midY = SCREEN_HEIGHT / 2;
display.drawFastHLine(0, midY, SCREEN_WIDTH, SSD1306_WHITE);

int start = (bufIndex + 1) % NUM_SAMPLES;
int prevX = 0;
int prevIdx = start;
float prevFraction = 0.0f;
float span = (float)(maxVal - minVal);
for (int i = 0; i < NUM_SAMPLES; ++i) {
    int idx = (start + i) % NUM_SAMPLES;
    float fraction = 0.0f;
    if (span != 0.0f) fraction = (samples[idx] - mid) / span;
    int x = i;
    int y = midY - (int)roundf(fraction * (float)(SCREEN_HEIGHT / 2 - 8));
    y = constrain(y, 10, SCREEN_HEIGHT - 12);
    if (i > 0) display.drawLine(prevX, constrain(prevIdx, 10, SCREEN_HEIGHT - 12), x,
y, SSD1306_WHITE);
    prevX = x;
    prevIdx = y;
}

display.setCursor(0, 54);
display.printf("Freq: %.1fHz  DC: %.2fV", freq, dc);
if (paused) {
    display.setCursor(90, 0);
    display.print("[PAUSED]");
}
display.display();
}

```

CHAPTER - 6

TESTING AND RESULTS

The portable oscilloscope operates by integrating multiple hardware and software components to capture, process, and display electrical waveforms in real time. The testing phase involved verifying the accuracy of the ADS1115 ADC, the processing capability of the ESP32, and the graphical response of the OLED display. Input signals of different types—such as sine, square, and triangular waves—were supplied using a function generator. These signals were sampled by the ADS1115, which converted the analog voltages into high-precision digital data. The ESP32 processed these samples by mapping them to the corresponding pixel positions on the OLED display, allowing the user to observe waveform shapes clearly on the screen.

During testing, various signal parameters such as **peak-to-peak voltage (Vpp)**, **frequency**, and **amplitude variations** were computed and displayed in real time. The system's sampling logic and timing functions were refined to ensure smooth waveform plotting without flickering. The OLED display successfully rendered the incoming waveform in a scrolling graph format, updating continuously as new samples arrived from the ADC. This provided a real-time graphical representation similar to that of a basic laboratory oscilloscope.

To evaluate performance, signals of different frequencies were applied. At lower and moderate frequencies, the oscilloscope displayed smooth, stable waveforms with accurate amplitude representation. Frequency calculations were validated by comparing measured values on the OLED with those displayed on the function generator, confirming the correctness of the zero-crossing detection algorithm implemented in the ESP32. Peak-to-peak voltage readings were also measured and found to be consistent with theoretical values, demonstrating the accuracy of the ADS1115 ADC.

The hardware setup—including the ESP32, ADS1115, and OLED display—was mounted on a prototyping board and tested for stability, noise immunity, and power consumption. The system responded effectively to rapid changes in input signals,

maintaining continuous waveform updates without significant delays. The portability and compact size of the device made it convenient for bench-top and field testing.

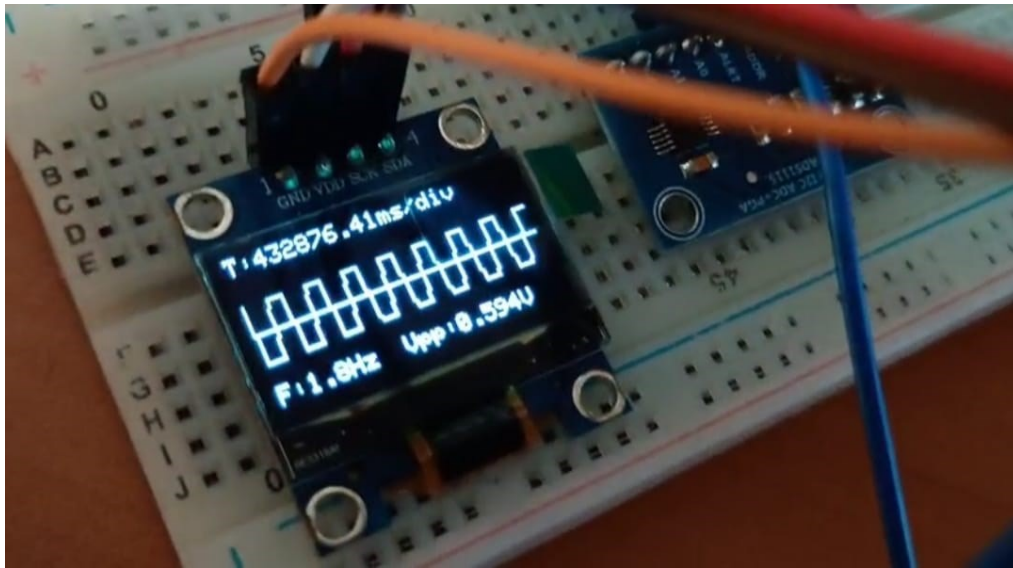


Fig 6.1: REAL-TIME WAVEFORM DISPLAYED ON OLED

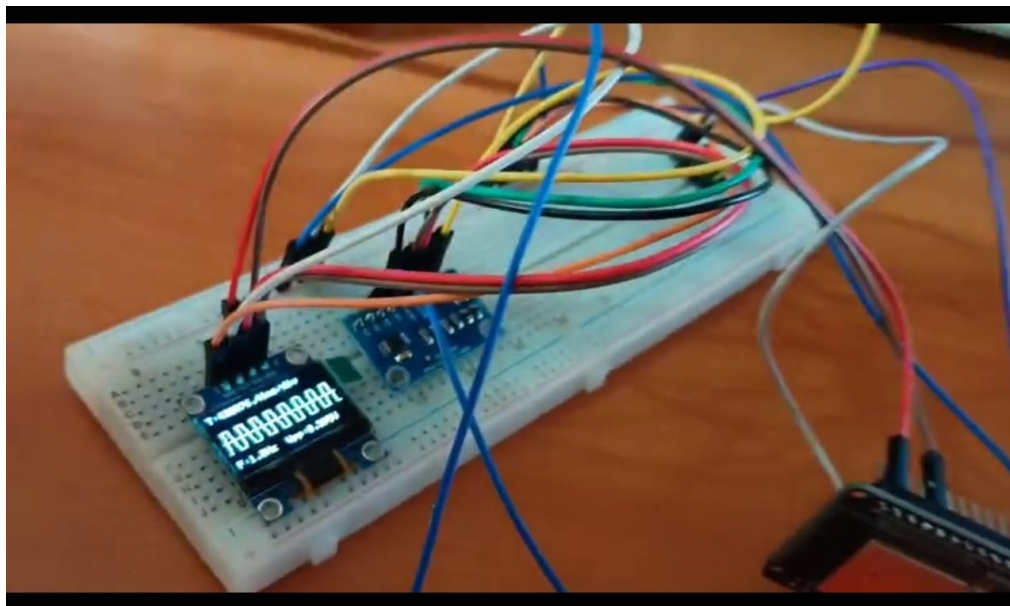


Fig 6.2: HARDWARE SETUP OF THE PORTABLE OSCILLOSCOPE

The results show that the portable oscilloscope successfully replicates essential functionalities of a basic signal-analysis instrument. It is capable of capturing, processing, and displaying real-time waveforms, making it suitable for educational use, electronics prototyping, and small-scale diagnostics. The accurate rendering of waveforms and signal parameters validates the effectiveness of the design and demonstrates the feasibility of creating a low-cost, compact oscilloscope using ESP32 and ADS1115.



Fig 6.3: GROUP PHOTO WITH GUIDE

CHAPTER - 7

FUTURE ENHANCEMENTS

The portable oscilloscope developed in this project demonstrates the feasibility of creating a compact, cost-effective, and user-friendly diagnostic tool for real-time signal monitoring. Although the current design successfully captures and displays essential waveform characteristics such as amplitude, peak-to-peak voltage, and frequency, there is considerable potential for future development to enhance its performance, usability, and application scope.

One of the primary enhancements involves increasing the **sampling rate and bandwidth** of the system. By integrating higher-speed ADCs or optimizing ESP32 processing algorithms, the oscilloscope can be improved to support higher-frequency signal analysis, enabling use beyond basic educational or low-frequency applications. Expanding the system to support **multi-channel signal acquisition** would also provide users the ability to compare two or more signals simultaneously, similar to commercial oscilloscopes.

Another significant enhancement lies in incorporating **wireless connectivity and data logging**. ESP32's built-in Wi-Fi/Bluetooth can be utilized to transmit waveform data to mobile apps or cloud platforms for advanced visualization, long-term storage, and remote monitoring. This allows users to analyze waveforms on smartphones or computers and maintain historical signal records for research or diagnostics.

Additional improvements may include implementing **FFT (Fast Fourier Transform)** for frequency-domain analysis, adding a **larger display** for improved waveform clarity, integrating rechargeable battery power for true portability, and providing **automatic scaling and triggering options** to enhance signal stability. These enhancements would not only elevate the oscilloscope's functionality but also broaden its use cases in laboratories, field environments, and academic institutions.

By incorporating such advancements, the portable oscilloscope can evolve into a more sophisticated and versatile instrument, comparable to entry-level commercial oscilloscopes while still maintaining affordability and ease of use.

7.1 CONCLUSION :

The Portable Oscilloscope using ESP32 successfully demonstrates how modern microcontrollers, high-precision ADCs, and compact display technologies can be combined to create an efficient and accessible signal analysis tool. The system provides real-time waveform visualization and essential measurement features such as frequency and peak-to-peak voltage, making it highly suitable for students, electronics hobbyists, and technicians who require a lightweight and portable diagnostic instrument.

Compared to traditional oscilloscopes, which are often expensive and bulky, this portable solution offers a practical alternative for basic signal monitoring tasks. The project validates the capability of ESP32 to process sampled data effectively and display waveforms smoothly on an OLED screen, while the ADS1115 ADC ensures accurate analog-to-digital conversion. The modular design of the system allows easy customization, making it adaptable to various embedded applications.

In summary, this project highlights the potential of low-cost embedded platforms to deliver meaningful test and measurement functionality. It serves as a valuable educational tool for understanding waveform behavior, signal parameters, and real-time digital processing. With further enhancements, the system can evolve into a more advanced portable oscilloscope, expanding its scope in academic, industrial, and research environments.

7.2 REFERENCES

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